SCHWARTZ - ZIPPEL LEMMA

NOTATION :

Given a polynomial $p(x_1, \dots x_n)$,

$$Z(p) = \{(x_1, \dots x_n) \mid p(x_1, \dots x_n) = 0\}$$

$d_i = \deg_{x_i} p$ is the degree of x_i in p

$p(x, y, z) = x^3 + xy + z^4 y$

$D = \deg p$ is the degree of p

Lemma 1. *Suppose that Q is a polynomial in the variables $x_1, \ldots, x_n$ and that Q is not identically zero. Let Q_1 be the standard simplified form of Q. Let d_1 be the degree of Q_1 in x_1 and Q_2 the coefficient of $x_1^{d_1}$ in Q_1. Then, inductively, let d_j be the degree of Q_j in x_j and Q_{j+1} the coefficient of $x_j^{d_j}$ in Q_j. This defines d_j and Q_j for $j = 1, \ldots, n$. For $1 \le j \le n$, let I_j be any set of elements in the domain or field F of coefficients of Q. Then in the set $I_1 \times \cdots \times I_n$, Q has at most*

$$|I_1 \times \cdots \times I_n| \left(\frac{d_1}{|I_1|} + \cdots + \frac{d_n}{|I_n|} \right) \tag{1}$$

zeros.

$$\| \qquad |Z(Q) \cap I_1 \times \cdots \times I_n|$$

$$p(x,y,z) = x^3 + yx^2 + zy^2 + zx^2 + zx + z^3$$

$$x^3 + (y+z)x^2 + (z)\cdot x + (zy^2 + z^2 + z^3)$$

$$p(x,y,z) = x^3y^2z^3 + x^3y^2z^4 + x^2y$$

$$Q_1(x,y,z) = x^3\underbrace{(y^2z^3 + y^2z^4)}_{Q_2(y,z)} + x^2y^2$$

$$Q_2(y,z) = y^2\underbrace{(z^4 + z^3)}_{Q_3(z)}$$

$d_1 = 3$

$d_2 = 2$

$d_3 = 4$

Corollary 1. Let $I = I_1 = \cdots = I_n$, and let $|I| \ge c\,\deg(Q)$. Then if Q is not identical to zero, the number of elements of $I \times \cdots \times I$ which are zeros of Q is at most $c^{-1}|I|^n$.

$$\frac{\deg(Q)}{|I|} \le \frac{1}{c}$$

PROOF : $P(x_1, \ldots, x_n) = \sum_{i=1}^{d_1} x_1^{\,i}\, P_2^{\,i}(x_2, \ldots x_n)$

By induction :

→ if $n=1$, trivial, a non-zero polynomial of degree d could have at most d zeros $|Z(p) \cap I| = d = |I| \cdot \left(\frac{d}{|I|}\right)$

→ if the formula hold for every polynomials in $n-1$ variables

$$Q_1 = x_1^{d_1} \cdot Q_2 + x_1^{d_1 - 1} \cdot \tilde{Q}_2 + \cdot$$

if $Q_2 = 0$ → at most $d_1 - 1 \le I_1$

$Q_2 \neq 0$ → at most d_1

$$|Z(Q_1) \cap I_1 \times \cdots I_n| \le$$

$$\le |I_1| \cdot |Z(Q_2) \cap I_2 \times \cdots \times I_n| + d_1 \cdot |I_2 \times \cdots \times I_n| =$$

$$= |I_1| \cdot |I_2 \times \cdots \times I_n| \cdot \left(\frac{d_2}{|I_2|} + \cdots + \frac{d_n}{|I_n|}\right) + d_1 \cdot |I_2 \times \cdots \times I_n| =$$

$$= |I_1 \times \cdots \times I_n| \cdot \left(\frac{d_2}{|I_2|} + \cdots + \frac{d_n}{|I_n|}\right) + \frac{d_1}{|I_1|} \cdot |I_1 \times \cdots \times I_n| =$$

$$= |I_1 \times \cdots \times I_n| \left(\frac{d_1}{|I_1|} + \cdots + \frac{d_n}{|I_n|}\right)$$

PROOF : From Lemma 1, we know

$$|Z(Q_1) \cap I_1 \times \cdots \times I_n| \le$$

$$\le |I_1 \times \cdots \times I_n| \left(\frac{d_1}{|I_1|} + \cdots + \frac{d_n}{|I_n|}\right)$$

$$= |I|^n \cdot \frac{d_1 + \cdots + d_n}{|I|} \le$$

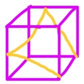
$$\le |I|^n \cdot \frac{\deg(Q_1)}{|I|} \le \frac{|I|^n}{c}$$

Lemma 2. *Let the hypotheses of Lemma 1 be satisfied. Suppose in addition that the coefficients of Q are integers, that $I = I_1 = I_2 = \cdots = I_n = \{i \mid -k \le i \le k\}$, and that $L = \max v(Q, k)$. Let J be any finite set of primes, and suppose that the product of any $m + 1$ of the primes in J exceeds L. Then in the set $I_1 \times \cdots \times I_n \times J$, Q has at most*

$$|I_1 \times \cdots \times I_n \times J| \left(\frac{d_1}{|I_1|} + \cdots + \frac{d_n}{|I_n|} + \frac{m}{|J|} \right) \tag{2}$$

modular zeros.

$$\max_{-k \le x_i \le k} \{Q(x_1, \ldots, x_n)\}$$



$(x_1, \ldots x_n, p)$ integers

$Q(x_1, \ldots x_n) \equiv 0 \mod p$

EXAMPLE : $J = \{2, 3, 5, 7\}$

$m = 2$

$L = 29$

every product of 3 prime numbers in J exceeds L

PROOF : We want to bound the number of modular zeros

$Q(x_1, \ldots x_n) = 0$ → from lemma 1 we have a bound

$Q(x_1, \ldots x_n) = k \cdot p_i$

$$|Q(x_1, \ldots x_n)| = q \le L \le p_1 \cdots p_{m+1}$$

The value 'q' could be the product of at most 'm' prime numbers

$$|Z^{MOD}(Q) \cap I_1 \times \cdots I_n \times J| \le$$

$$\le |J| \cdot |I_1 \times \cdots \times I_n| \left(\frac{d_1}{|I_1|} + \cdots + \frac{d_n}{|I_n|}\right) + m \cdot |I_1 \times \cdots \times I_n|$$

$$\le |J| \cdot |I_1 \times \cdots \times I_n| \left(\frac{d_1}{|I_1|} + \cdots \times \frac{d_n}{|I_n|}\right) + |J| \cdot \frac{m}{|J|} \, |I_1 \times \cdots \times I_n| =$$

$$= |J| \cdot |I_1 \times \cdots \times I_n| \cdot \left(\frac{d_1}{|I_1|} + \cdots + \frac{m}{|J|}\right)$$

$$= |I_1 \times \cdots \times I_n \times J| \cdot \left(\frac{d_1}{|I_1|} + \cdots + \frac{m}{|J|}\right)$$

Corollary 2. *Let $2k + 1 \ge c \cdot \deg(Q)$, and suppose that the product of the $c^{-1}|J| + 1$ smallest primes in J exceeds $\max v(Q, k)$. Then if Q is not identically equal to zero, the number of elements of $I_1 \times \cdots \times I_n \times J$ which are modular zeros of Q is at most $2c^{-1}|I|^n|J|$.*

$$m = \frac{|J|}{c}$$

PROOF : From Lemma 2

$$|Z^{MOD}(Q) \cap I_1 \times \cdots \times I_n \times J| \le$$

$$\le |I_1 \times \cdots \times I_n \times J| \left(\frac{d_1}{|I_1|} + \cdots + \frac{d_n}{|I_n|} + \frac{m}{|J|}\right) \le$$

$$\le |I|^n \cdot |J| \cdot \left(\frac{d_1 + \cdots + d_n}{|I|} + \frac{|J|}{|J|c}\right) \le$$

$$\le |I|^n \cdot |J| \cdot \left(\frac{\deg(Q)}{|I|} + \frac{1}{c}\right) \le$$

$$\le |I|^n \cdot |J| \left(\frac{2k+1}{c\,|I|} + \frac{1}{c}\right) \le$$

$$\Big\downarrow |I| = 2k+1$$

$$= |I|^n \cdot |J| \left(\frac{1}{c} + \frac{1}{c}\right) = 2\,\frac{|I|^n \cdot |J|}{c}$$

Zip 79

Theorem 1. *Let $f \in \mathbb{Z}[X_1, \dots, X_v]$ and the degree of f in X_i be bounded by D. Let $N_v(B)$ be the number of zeroes of f, $(x_1, \dots, x_v)$ such that $x_i \in \mathfrak{I}$ (a set with B elements, $B \gg D$). Then $N_v(B) \leq B^v - (B-D)^v$.*

Given $Q(x_1, \dots x_n)$, $\deg_{x_i} Q \leq D$

$$|Z(Q) \cap I^n| \leq |I|^n - (|I|-D)^n$$

PROOF: $P(x_1, \dots, x_n) = \sum_{i=1}^{d_1} x_1^i P_2^i(x_2, \dots x_n)$

By induction:

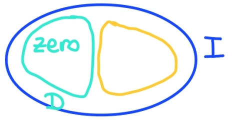
→ if $n=1$, trivial, a non-zero polynomial of degree d could have at most d zeros

$|Z(p) \cap I| = d = |I| - (|I| - d)$

→ if the formula hold for every polynomials in $n-1$ variables

$$Q_1 = x_1^{d_1} \cdot Q_2 + x_1^{d_1 - 1} \cdot \tilde{Q}_2 + \cdot$$

for each $x_2, \dots x_n$ we can split the set I for the first variable into two disjoint set



$$|Z(Q) \cap I^n| \leq$$

$$\leq \boxed{D \cdot |I|^{n-1}} + \boxed{(|I|-D) \cdot (Z(Q_2) \cap I^{n-1})} =$$

$$\leq D \cdot |I|^{n-1} + (|I| - D) \cdot (|I|^{n-1} - (|I|-D)^{n-1}) =$$

$$\leq D \cdot |I|^{n-1} + |I|^n - |I| \cdot (|I|-D)^{n-1} - D \cdot |I|^{n-1} + D(|I|-D)^{n-1} =$$

$$\leq |I|^n - (|I|-D)(|I|-D)^{n-1} =$$

$$\leq |I|^n - (|I|-D)^n$$

Lemma 1.6 (The Schwartz-Zippel Lemma). *Let $p(x_1, \dots, x_n)$ be a nonzero polynomial of n variables with degree d. Let S be a finite subset of $\mathbb{R}$, with at least d elements in it. If we assign $x_1, \dots, x_n$ values from S independently and uniformly at random, then*

$$\mathbb{P}[p(x_1, \dots, x_n) = 0] \leq \frac{d}{|S|}.$$

in the previously used notation

$$\mathbb{P}[Q(x_1, \dots x_n) = 0] \leq \frac{D}{|I|} \qquad \forall (x_1, \dots x_n) \xleftarrow{\$} I$$

PROOF: $$\mathbb{P}[Q(x_1, \dots, x_n) = 0] = \frac{|Z(Q) \cap S|}{|S|^n} \leq \frac{|S|^n \cdot \frac{D}{|S|}}{|S|^n} = \frac{D}{|S|}$$

Applications

GRAPH

POLYNOMIAL IDENTITY (PIT)

$P_1 = P_2 \implies Q = P_1 - P_2$

IDEA: probabilistic test to check if a polynomial is identically zero

$Z_i = Q(r_i) = 0$

$\mathbb{P}(Z_i) \leq \frac{d}{|S|}$ } this is the probability that a random evaluation is zero

If we repeat the test several times:

$\mathbb{P}(Z_1 \cap \dots Z_k) = \prod(\mathbb{P}(Z_i)) = \left(\frac{d}{|S|}\right)^k$

LINEAR ALGEBRA

Matrix identity test

$AB \stackrel{?}{=} C$ $\} \sim O(n^3)$

$\downarrow$

$AB \cdot r - C \cdot r = 0$

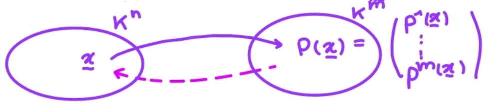
$R_i =$ " $AB \cdot r_i - C \cdot r_i = 0$ "

$\mathbb{P}(R_i) = \frac{1}{|S|}$

$\mathbb{P}(R_1 \cap \dots \cap R_k) = \frac{1}{|S|^k}$

CRYPTOGRAPHY

- sumcheck interactive protocols
- zk-SNARK
- MQ problem:
  Multivariate Quadratic problem:
  (*) given a MQ-polynomial
  $p^1(x_1, \dots x_n) = \sum a_{ij} x_i x_j + \sum b_i x_i + c$
  $P(x) = (p^1(x), \dots p^m(x))$



MQ on my mind - MQOM - based on MPC-in-the-head paradigm to build a zero knowledge proof of knowledge for MQ. (Euro S&P 24)

$[\![x]\!]$ $\begin{cases} p_1(x) \\ \vdots \\ p_m(x) \end{cases} \longrightarrow \sum \gamma_i \, p_i(x) \longrightarrow$ can cheat with error bound by $\frac{1}{q^n}$ on field $\mathbb{F}_{q^n}$

UOV - unbalanced oil and vinegar scheme

$p(x_1, \dots x_n)$

$\downarrow$

$p(v_1, \dots v_m, o_1, \dots o_{n-m}) = \sum a_{ij} v_i v_j + \sum b_{ij} v_i o_j + \sum c_i v_i + \sum d_i o_j + c$

$p(\bar{v}, o) = \sum a_{ij} v_i v_j + \sum b_{ij} v_i o_j + \sum c_i v_i + \sum d_i o_j + c$